

Multigrid Finite State Projection for the Reaction-Diffusion Master Equation

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The Chemical Master Equation

Integer counts $n \in \mathbb{Z}_{\geq 0}$ evolving by stochastic reactions. The probability distribution $p(n, t)$ satisfies:

$$\dot{p} = A p$$

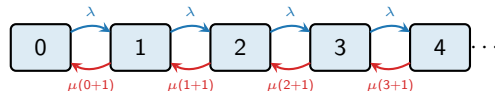
SSA vs. CME

SSA samples individual trajectories.

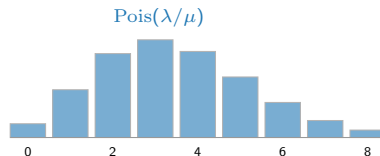
CME gives the *exact* joint distribution $p(n, t)$ — rare events, tails, bimodal structure, no ensemble noise.

FSP: truncate A to a finite support $\mathcal{S}$, solve $\dot{p} = A p$ via expv. Expand $\mathcal{S}$ at the boundary; prune negligible states.

Birth-death: $\emptyset \xrightarrow{\lambda} X \xrightarrow{\mu n} \emptyset$



Stationary: $\text{Pois}(\lambda/\mu)$.



The RDME: exact but exponentially large

Divide the domain into $K \times K$ voxels. Each voxel $\mathbf{k}$ holds count $n_{\mathbf{k}}$.

Two event types:

- **Reactions** inside $\mathbf{k}$: propensity $\alpha(n_{\mathbf{k}})$
- **Diffusion** $\mathbf{k} \leftrightarrow \mathbf{l}$: rate $d=D/h^2$, one molecule

Joint distribution $p(\{n_{\mathbf{k}}\}, t)$ satisfies $\dot{p} = Ap$ (RDME).

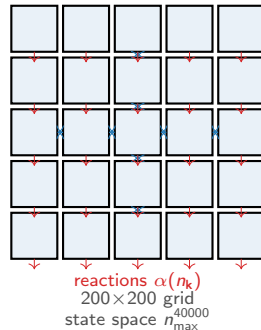
The curse of dimensionality

State space: $n_{\max}^{K^2}$.

A 200×200 grid is **completely intractable** directly.

Key locality: diffusion only couples *adjacent* voxels
— this structure enables a principled decomposition.

diffusion $d=D/h^2$



Three voxel regimes: adaptive classification at each step

Every voxel $\mathbf{k}$ belongs to one of three regimes, reassigned each step:

Active (wavefront)

Full CME distribution $p_k(n)$ tracked.

Evolved by the multigrid V-cycle — captures joint dynamics of neighbouring voxels.

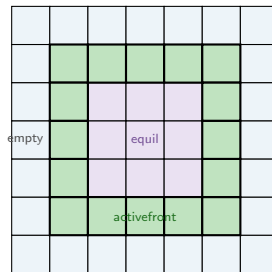
Typically a thin ring around the wavefront.

Equilibrated (interior/exterior)

Full distribution $p_k(n)$ tracked.

Evolved cheaply by a **Galerkin-reduced single-voxel CME**:

$$Q_k^{\text{eff}}[n \rightarrow n+1] = r_b + d \sum_{\ell \sim k} \mu_\ell$$



Key: equil voxels track the *full* distribution (not just the mean), so cross-basin events and non-Poisson tails are preserved.

The multigrid V-cycle (next slide) handles *only* the thin active front — where inter-voxel correlations matter most.

Spatial multigrid V-cycle: three levels, one time step

Hierarchy over active voxels:

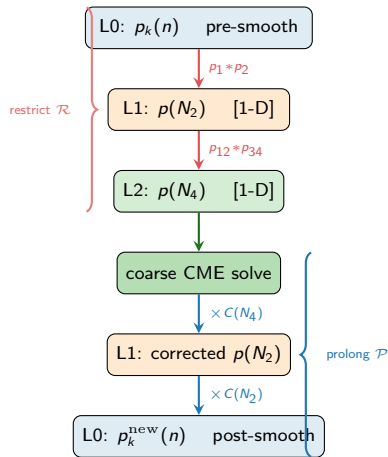
- **L0** individual voxels: $p_k(n)$, w states
- **L1** adjacent pairs: $p(N_2=n_1+n_2)$, 1-D
- **L2** 2×2 quads: $p(N_4)$, 1-D [coarsest]

One V-cycle = exactly Δt :

$$\alpha_{\text{pre}}\Delta t + (1-2\alpha)\Delta t + \alpha_{\text{post}}\Delta t = \Delta t$$

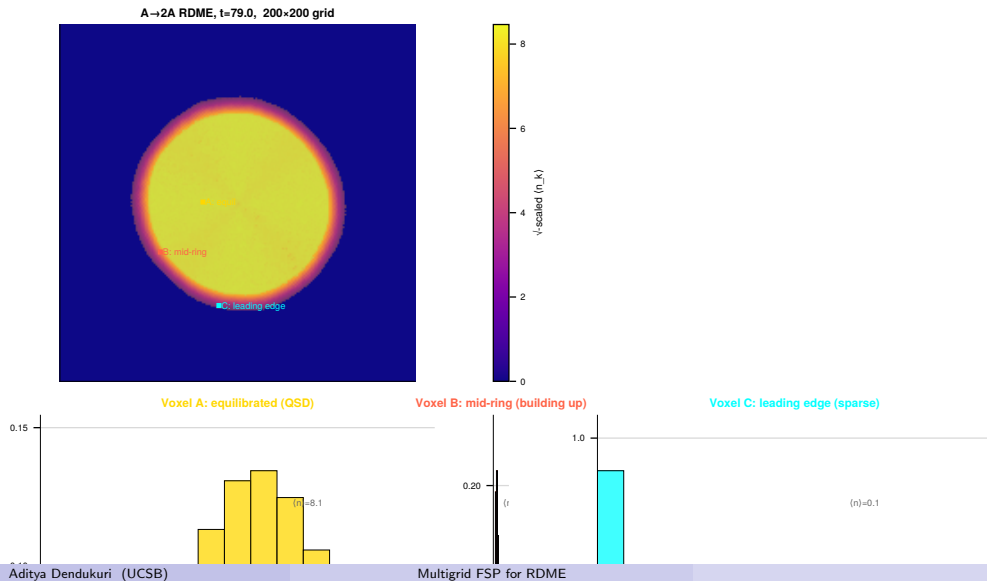
- 1 **Pre-smooth** L0: per-voxel expv $[\alpha_{\text{pre}}\Delta t]$
- 2 **Restrict** L0 $\rightarrow$ L1: $p_{12}(N_2) = p_1 * p_2$
- 3 **Restrict** L1 $\rightarrow$ L2: $p_{1234}(N_4) = p_{12} * p_{34}$
- 4 **Coarse solve** L2: 1-D CME $[(1-2\alpha)\Delta t]$
- 5 **Correct** L2 $\rightarrow$ L1: $p_{12}^{\text{new}} \propto p_{12} \times E[C(N_4) | N_2]$
- 6 **Correct** L1 $\rightarrow$ L0: $p_k^{\text{new}} \propto p_k \times E[C(N_2) | n_k]$
- 7 **Post-smooth** L0: per-voxel expv $[\alpha_{\text{post}}\Delta t]$

Multiplicative correction at each level:



ADI: alternate row/col pairs each step
captures all 4 nearest-neighbour correlations

Result: $A \rightarrow 2A$ wavefront on a 200×200 RDME



Bistable Schlögl RDME: two stable basins

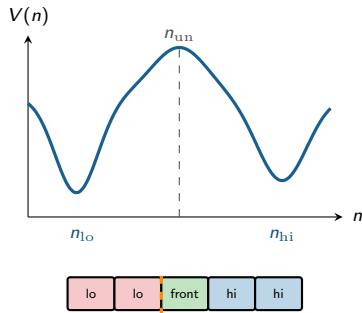
Schlögl model: $2X \rightleftharpoons 3X$, $\emptyset \rightleftharpoons X$. Two stable fixed points: $n_{lo} \approx 38$, $n_{hi} \approx 164$.

Voxel classification for bistability

Equilibrated-lo/hi: distribution frozen as a delta at n_{lo} or n_{hi} . Re-activated only when wavefront flux crosses the unstable point n_{un} .

Active front: full CME distribution spanning both basins, evolved by the multigrid V-cycle.

No special-casing: the same algorithm handles the bistable case natively.

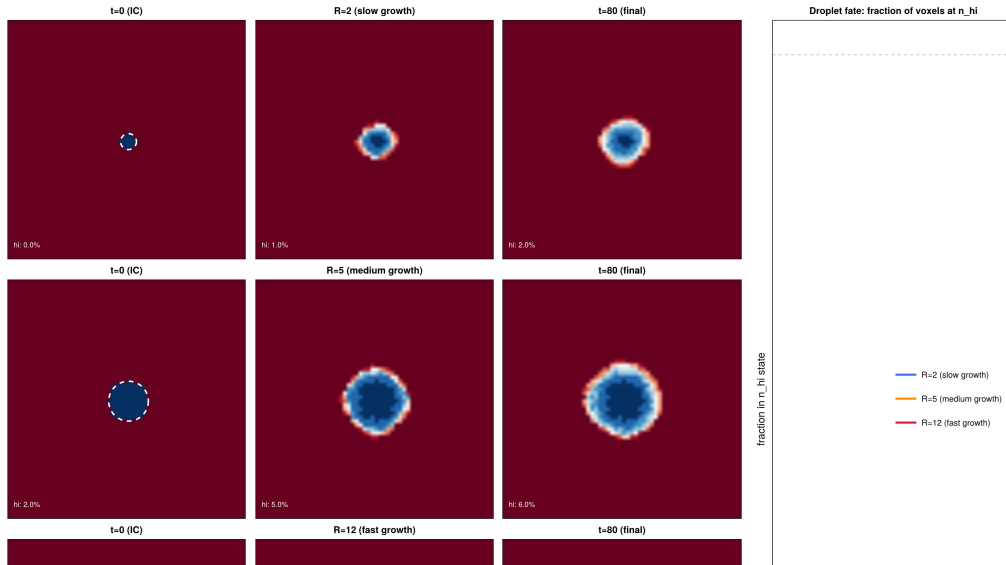


$P(hi)$ colormap

Color each voxel by $P(n > n_{un})$:

red = equil-lo, **blue** = equil-hi, **white** = transitioning front

Result: Schlögl critical droplet on a 60×60 grid

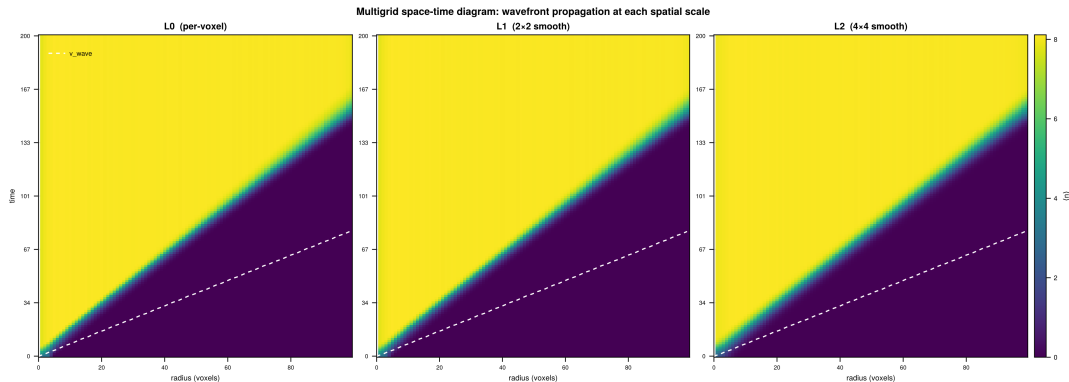


The correction field: L2 adds information at the front



200×200 BD RDME at peak wavefront ($t=142$, 4138 active voxels).

Space-time diagram: each level sees the wavefront differently



Heatmap: radial mean $\langle n \rangle$ vs. radius (x) and time (y). White dashed: theoretical $v \approx 1.26$.

L0 (per-voxel): sharp thin band. **L1** (2-voxel smooth): slightly broader front. **L2** (4-voxel smooth): broadest — quads capture collective dynamics *ahead* of the fine-level front.

Non-redundancy across levels is why the hierarchy works: each level carries distinct spatial-scale information that the others cannot resolve.

Summary

1. Adaptive voxel classification with Galerkin-reduced CME.

The full RDME ($n_{\max}^{K^2}$ states) is never assembled. **Equilibrated** voxels track the full distribution, evolved cheaply via a single-voxel Galerkin CME with effective rates from neighbour means. **Empty** voxels are activated organically as the wavefront advances.

2. Spatial multigrid V-cycle at the active front.

Three-level hierarchy: L0 (voxels) $\rightarrow$ L1 (pairs) $\rightarrow$ L2 (quads). Restriction by convolution; coarse CME solve captures 4-voxel collective dynamics; multiplicative correction propagates L2 $\rightarrow$ L1 $\rightarrow$ L0. Time is split exactly: $\alpha_{\text{pre}} + (1-2\alpha) + \alpha_{\text{post}} = \Delta t$.

$\sim 40\%$ speedup on 200×200 BD wavefront. Correction field confirms L2 acts only at the active front.

3. Model-agnostic: handles bistability natively.

Schlögl equil voxels are frozen at their basin (n_{lo} or n_{hi}); re-activated only when wavefront flux crosses n_{un} . The multigrid evolves only the thin transitioning front. Circular droplets on 60×60 with correct P(hi) dynamics.